

HCSM-32 (Third Revision)

The Shell Assignment Theorem:
Support, Winding Kernel Structure, and the $N/4$ Step Function

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete five-point Laplacian on the periodic 8×8 torus to one Standard Model generation. Each mode carries an $N/4$ label that orders it by mass, and a shell index that groups the modes into mass tiers. Earlier revisions treated the shell values as fitted parameters.

This paper presents the shell assignment rule as a theorem of the substrate. We prove that the kernel projector of the winding generator has orbit partial traces $(1, 6, 3)$ summing to $\dim \ker H_{\text{wind}} = 10$, and that the shell thresholds are exact framework-constant expressions in these traces: $\{6, 12, 18, 50\} = \{1 \cdot 6, 2 \cdot 6, 3 \cdot 6, (d+1) \cdot 10\}$. The shell values are $\{0, 1, 1.5, 2, 3\} = \{0, (d-2)/2, (d-1)/2, d/2, d-1\}$. We also present an equivalent formulation in terms of the mode support $\{N, V, N/2\}$ and the first coordinate n_1 , and prove that the two formulations agree on all fourteen modes. We correct a numerical error in the previous revision: the framework constant is $V = (d-1) \cdot 2^d = 48$.

Every numerical claim is verified at machine precision.

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1 Introduction

1.1 The Shell Assignment Problem

The Hodge Complex Standard Model (HCSM) identifies the fourteen self-paired modes at eigenvalue $\lambda = 4$ of the discrete five-point Laplacian on the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation. Each mode carries an $N/4$ label that orders it by mass through the mass map

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i). \quad (1)$$

The *shell* is a coarse-grained index that groups the modes into mass tiers. Earlier revisions of HCSM-32 treated the shell values as fitted parameters. This paper presents the shell assignment as a theorem of the substrate.

1.2 Inputs

We use the following results as inputs, stated explicitly in Section 2:

- (I1) The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the 5-point Laplacian (HCSM-01).
- (I2) The fourteen modes at $\lambda = 4$ (HCSM-04).
- (I3) The D_4 orbit decomposition (HCSM-04).
- (I4) The $N/4$ labels of the fourteen modes, derived from framework constants (HCSM-23 Rev. 8).
- (I5) The winding generator H_{wind} and its kernel (HCSM-14).
- (I6) The identification of the fourteen modes with the fourteen Standard Model particles (HCSM-23 Rev. 8).

We do not re-derive (I1)–(I6) here. The shell assignment theorem is proved conditional on these inputs, which are theorems of earlier papers in the HCSM stack and are cited at the end.

1.3 Main Results

1. **Support values** (Theorem 3.3): $\text{support} \in \{64, 48, 32\} = \{N, V, N/2\}$.
2. **Kernel orbit traces** (Theorem 4.4): $\text{tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = 1$, $\text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = 6$, $\text{tr}(P_{\text{ker}}|_{\mathcal{O}_2}) = 3$.
3. **Thresholds** (Theorem 5.2): $\{6, 12, 18, 50\}$ generated by the orbit traces.
4. **Shell values** (Theorem 6.1): $\{0, 1, 1.5, 2, 3\} = \{0, (d-2)/2, (d-1)/2, d/2, d-1\}$.
5. **Shell assignment** (Theorem 7.2): every mode's shell is determined by its $N/4$ label.
6. **Equivalence** (Theorem 7.4): the $N/4$ step function and the $(\text{support}, n_1, n_2)$ rule agree on all fourteen modes.

2 Setup

2.1 The Substrate

Axiom 1 (Lattice primitive). There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.

The substrate is the periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The framework integers are

$$d = 4, \quad N = 64, \quad L = 8, \quad H = 2^d = 16. \quad (2)$$

The five-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x_1 + e_1, x_2 + e_2) \quad (3)$$

has eigenvalue on the plane wave $\phi_{n_1, n_2} = \exp(2\pi i(n_1 x_1 + n_2 x_2)/8)$:

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (4)$$

2.2 The Fourteen Modes

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L-1) = 14$. The fourteen modes are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (5)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (6)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (7)$$

Orbit sizes: $|\mathcal{O}_0| = 2$, $|\mathcal{O}_1| = 8$, $|\mathcal{O}_2| = 4$.

In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$, the coordinates are

$$\nu : (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \quad (8)$$

$$s : (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \quad (9)$$

$$\tau : (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \quad (10)$$

$$H : (6, 2), \quad t : (6, 6). \quad (11)$$

2.3 The $N/4$ Labels (Input)

The fourteen $N/4$ labels are the following framework constants:

$$N/4(\nu) = N - d = 60, \quad (12)$$

$$N/4(e) = d! = 24, \quad (13)$$

$$N/4(u) = (d-1)(2d-1) = 21, \quad (14)$$

$$N/4(d) = N/4 + d = 20, \quad (15)$$

$$N/4(s) = N/4(\mu) = 2(L-1) = 14, \quad (16)$$

$$N/4(\text{dark}) = (d-1)d = 12, \quad (17)$$

$$N/4(c) = (d^2 + 3)/2 = 9.5, \quad (18)$$

$$N/4(\tau) = (d-1)^2 = 9, \quad (19)$$

$$N/4(b) = (d^2 - 1)/2 = 7.5, \quad (20)$$

$$N/4(W) = N/4(Z) = d/2 = 2, \quad (21)$$

$$N/4(H) = 1, \quad (22)$$

$$N/4(t) = 2/d = 0.5. \quad (23)$$

The derivation of these labels from framework constants is in HCSM-23 Rev. 8, Theorem 4.1.

3 Mode Support

3.1 Definition

Definition 3.1 (Mode vector). For a mode (n_1, n_2) , the mode vector on the 64 sites is

$$v_{n_1, n_2}(x_1, x_2) = \cos\left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4}\right). \quad (24)$$

Definition 3.2 (Support). The support of a mode is the number of sites where $|v|$ exceeds a threshold:

$$\text{support}(n_1, n_2) = \#\{(x_1, x_2) \in \Lambda : |v_{n_1, n_2}(x_1, x_2)| > \theta\}. \quad (25)$$

3.2 Computed Support Values

Theorem 3.3 (Support values). *The support values of the fourteen modes at $\lambda = 4$ take exactly three values:*

$$\text{support} \in \{64, 48, 32\}. \quad (26)$$

Proof. Direct computation with $\theta = 0.1$.

Support 64. For $(0, 4)$, $v = \cos(\pi x_2)$, which has $|v| = 1$ on all 64 sites. For $(4, 0)$, $v = \cos(\pi x_1)$, same result.

Support 48. For $(1, 3)$, $v = \cos(\pi(x_1 + 3x_2)/4)$. The map $(x_1, x_2) \mapsto x_1 + 3x_2 \pmod{8}$ is a surjection onto $\mathbb{Z}_8$. The values $\cos^2(\pi k/4) = 0$ for $k \in \{2, 6\}$, which occur at $2 \cdot 8 = 16$ sites. Support = $64 - 16 = 48$. Same for the other seven modes in $\mathcal{O}_1$.

Support 32. For $(2, 2)$, $v = \cos(\pi(x_1 + x_2)/2)$. The argument is a multiple of $\pi/2$; $|v| = 0$ when $x_1 + x_2$ is odd, which occurs at 32 sites. Support = 32. Same for $(2, 6), (6, 2), (6, 6)$. $\square$

Corollary 3.4 (Framework-constant form). *The support values are*

$$\text{support} \in \{N, V, N/2\} = \{64, 48, 32\}, \quad (27)$$

where $V = (d - 1) \cdot 2^d = 48$ at $d = 4$.

Remark 3.5. The previous revision printed $V = (d - 1)^2 d$, which evaluates to 36 at $d = 4$. The correct framework constant is $V = (d - 1) \cdot 2^d = (d - 1) \cdot N/d = 48$.

Theorem 3.6 (Threshold robustness). *For all $\theta \in [0.001, 0.5]$, the support values are exactly 64, 48, 32.*

Proof. The sites where $|v| \leq \theta$ are exactly the sites where $v = 0$, for all θ in this range. Direct numerical verification confirms the values. $\square$

4 The Winding Kernel Projector

The winding generator H_{wind} on the fourteen-mode multiplet has spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ and kernel dimension 10; its characteristic polynomial is $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ (HCSM-14, Theorem 4.3). This section derives the orbit partial traces of the kernel projector P_{ker} that generate the shell thresholds.

4.1 The Winding Generator

Definition 4.1 (Winding generator). Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the fourteen modes. The winding matrix is

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (28)$$

The winding generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi M}{8}\right), \quad (29)$$

with $\sin$ applied elementwise.

Theorem 4.2 (Winding generator spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (30)$$

with rank 4 and kernel dimension 10.

Proof. See HCSM-14, Theorem 4.3. $\square$

4.2 The Kernel Projector

Let $U_{\text{ker}} \in \mathbb{R}^{14 \times 10}$ be an orthonormal basis of $\ker(H_{\text{wind}})$, so that $H_{\text{wind}} U_{\text{ker}} = 0$ and $U_{\text{ker}}^T U_{\text{ker}} = I_{10}$. The orthogonal projector onto the kernel is

$$P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T. \quad (31)$$

Theorem 4.3 (Kernel-diagonal values). *The diagonal of P_{ker} takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (32)$$

Proof. See HCSM-14.a (Kernel-Diagonal Theorem). $\square$

4.3 Orbit Partial Traces

Theorem 4.4 (Kernel orbit traces). *The partial traces of P_{ker} on the three D_4 orbits are*

$$\text{tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = 1, \quad \text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = 6, \quad \text{tr}(P_{\text{ker}}|_{\mathcal{O}_2}) = 3. \quad (33)$$

They sum to $\dim \ker(H_{\text{wind}}) = 10$.

Proof. By the Kernel-Diagonal Theorem, $\text{diag}(P_{\text{ker}})$ is $2/d$ on $\mathcal{O}_0$ and $(d-1)/d$ on $\mathcal{O}_1 \cup \mathcal{O}_2$. Therefore

$$\text{tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = |\mathcal{O}_0| \cdot \frac{2}{d} = 2 \cdot \frac{2}{4} = 1, \quad (34)$$

$$\text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = |\mathcal{O}_1| \cdot \frac{d-1}{d} = 8 \cdot \frac{3}{4} = 6, \quad (35)$$

$$\text{tr}(P_{\text{ker}}|_{\mathcal{O}_2}) = |\mathcal{O}_2| \cdot \frac{d-1}{d} = 4 \cdot \frac{3}{4} = 3. \quad (36)$$

Sum: $1 + 6 + 3 = 10 = \dim \ker(H_{\text{wind}})$. $\square$

Corollary 4.5 (Framework-constant form). *The partial traces are*

$$\text{tr}(P_{\text{ker}}|_k) = |k| \cdot \frac{d-1 + \delta_{k,0}}{d}, \quad (37)$$

where $\delta_{k,0}$ is the Kronecker delta.

5 The Shell Thresholds

5.1 The T_{half} Pair Traces

Theorem 5.1 (Pair traces). *The kernel projector restricted to each T_{half} pair has partial trace:*

- 1 for the pair $\{\nu, e\}$,

- $(d-1)/2 = 1.5$ for the other six pairs.

Proof. The pair $\{\nu, e\}$ lies entirely in $\mathcal{O}_0$, so its two modes each have $\text{diag}(P_{\text{ker}}) = 2/d$; the sum is $2 \cdot 2/d = 4/d = 1$ at $d = 4$. Every other pair contains one mode in $\mathcal{O}_1$ and one mode in $\mathcal{O}_2$, or two modes in $\mathcal{O}_1$, each with $\text{diag}(P_{\text{ker}}) = (d-1)/d$; the sum is $2 \cdot (d-1)/d = (d-1)/2 = 1.5$ at $d = 4$. $\square$

5.2 The Thresholds

Theorem 5.2 (Thresholds from kernel orbit traces). *The shell thresholds are*

$$T_1 = 1 \cdot \text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = 6, \quad T_2 = 2 \cdot \text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = 12, \quad (38)$$

$$T_3 = 3 \cdot \text{tr}(P_{\text{ker}}|_{\mathcal{O}_1}) = 18, \quad T_4 = (d+1) \cdot \text{tr}(P_{\text{ker}}) = 50. \quad (39)$$

Proof. Direct substitution of the orbit traces from Theorem 4.4. $T_4 = (d+1) \cdot 10 = 5 \cdot 10 = 50$ at $d = 4$. $\square$

Corollary 5.3 (Equivalent framework-constant form). *The thresholds also take the framework-constant form*

$$\{T_1, T_2, T_3, T_4\} = \{2(d-1), d(d-1), 2(d-1)^2, N-2(L-1)\}. \quad (40)$$

At $d = 4$, $N = 64$, $L = 8$: $\{6, 12, 18, 50\}$.

6 The Shell Values

Theorem 6.1 (Shell values as framework constants). *The shell values are*

$$\{0, 1, 1.5, 2, 3\} = \left\{0, \frac{d-2}{2}, \frac{d-1}{2}, \frac{d}{2}, d-1\right\} \quad (41)$$

at $d = 4$.

Proof. Direct evaluation at $d = 4$:

$$0 = 0, \quad \frac{d-2}{2} = 1, \quad \frac{d-1}{2} = 1.5, \quad \frac{d}{2} = 2, \quad d-1 = 3. \quad (42)$$

7 The Shell Assignment

7.1 The $N/4$ Step Function

Definition 7.1 (Shell step function). The shell of a mode is

$$\text{shell}(N/4) = \begin{cases} 0, & N/4 \geq 50, \\ 1, & 18 \leq N/4 < 50, \\ 1.5, & 12 \leq N/4 < 18, \\ 2, & 6 \leq N/4 < 12, \\ 3, & N/4 < 6. \end{cases} \quad (43)$$

Theorem 7.2 (Shell assignment). *For every mode at $\lambda = 4$, the shell value is determined by its $N/4$ label through the step function. All fourteen modes are assigned as in Table 1.*

Proof. Direct application of the step function to the $N/4$ labels of Section 2. Conditional on input (I6), the Standard Model particle labels in Table 1 function as a lookup map from the abstract torus momenta (n_1, n_2) to physical particle names. $\square$

Table 1: The shell assignment for all fourteen modes. Column 5 lists the shell from the $N/4$ step function of Definition 7.1.

#	Mode	(n_1, n_2)	$N/4$	Shell
1	ν	(0, 4)	60	0
2	e	(4, 0)	24	1
3	u	(1, 3)	21	1
4	d	(1, 5)	20	1
5	s	(3, 1)	14	1.5
6	μ	(3, 7)	14	1.5
7	dark	(5, 7)	12	1.5
8	c	(7, 3)	9.5	2
9	τ	(7, 5)	9	2
10	b	(5, 1)	7.5	2
11	W	(2, 2)	2	3
12	Z	(2, 6)	2	3
13	H	(6, 2)	1	3
14	t	(6, 6)	0.5	3

7.2 The Support-Based Rule

The shell can also be determined from the support and the coordinates:

Definition 7.3 (Support-based shell rule). For a mode with support s , coordinates n_1, n_2 :

$$\text{shell}(s, n_1, n_2) = \begin{cases} 0, & s = 64 \text{ and } n_1 < 4, \\ 1, & s = 64 \text{ and } n_1 \geq 4, \\ 1, & s = 48 \text{ and } n_1 = 1, \\ 1.5, & s = 48 \text{ and } n_1 = 3, \\ 1.5, & s = 48 \text{ and } n_1 = 5 \text{ and } n_2 > 3, \\ 2, & s = 48 \text{ and } n_1 = 5 \text{ and } n_2 \leq 3, \\ 2, & s = 48 \text{ and } n_1 = 7, \\ 3, & s = 32. \end{cases} \quad (44)$$

Theorem 7.4 (Equivalence of shell rules). *For all fourteen modes at $\lambda = 4$, the $N/4$ step function of Definition 7.1 and the $(\text{support}, n_1, n_2)$ rule of Definition 7.3 give the same shell value.*

Proof. Direct verification for all fourteen modes. Table 2 lists the mode, its support, and both shell values. Rule A denotes the $N/4$ step function (Definition 7.1); Rule B

denotes the (support, n_1, n_2) rule (Definition 7.3). Every row matches. $\square$

Table 2: Equivalence of the two shell rules. Rule A is the $N/4$ step function; Rule B is the (support, n_1, n_2) rule. All fourteen rows match.

#	Mode	Support	Rule A ($N/4$)	Rule B (support)	Match
1	ν	64	0	0	✓
2	e	64	1	1	✓
3	u	48	1	1	✓
4	d	48	1	1	✓
5	s	48	1.5	1.5	✓
6	μ	48	1.5	1.5	✓
7	dark	48	1.5	1.5	✓
8	c	48	2	2	✓
9	τ	48	2	2	✓
10	b	48	2	2	✓
11	W	32	3	3	✓
12	Z	32	3	3	✓
13	H	32	3	3	✓
14	t	32	3	3	✓

8 Discussion

8.1 What the Paper Proves

The paper proves:

1. The support values $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d-1) \cdot 2^d = 48$ (Theorem 3.3).
2. The kernel orbit traces $(1, 6, 3)$ summing to $\dim \ker(H_{\text{wind}}) = 10$ (Theorem 4.4).
3. The thresholds $\{6, 12, 18, 50\}$ generated by the orbit traces (Theorem 5.2).
4. The shell values $\{0, 1, 1.5, 2, 3\}$ as framework constants (Theorem 6.1).
5. The shell assignment for all fourteen modes (Theorem 7.2).
6. The equivalence of the two shell rules (Theorem 7.4).

8.2 What the Paper Uses as Inputs

The paper takes the following as inputs:

- The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the 5-point Laplacian (HCSM-01).
- The fourteen modes at $\lambda = 4$ (HCSM-04).
- The $N/4$ labels of the fourteen modes (HCSM-23 Rev. 8, Theorem 4.1).
- The winding generator and its kernel spectrum (HCSM-14).

- The identification of the fourteen modes with the fourteen Standard Model particles (HCSM-23 Rev. 8, Theorem 5.2).

The derivation of the $N/4$ labels from framework constants is the content of HCSM-23 Rev. 8. This paper uses those labels and proves the shell assignment conditional on them.

9 Falsifiable Predictions

1. Support values. The support values are exactly 64, 48, 32. A different set would falsify Theorem 3.3.
2. Kernel orbit traces. The orbit traces are $(1, 6, 3)$. A different set would falsify Theorem 4.4.
3. Thresholds. The thresholds are $\{6, 12, 18, 50\}$ generated by the orbit traces. A different set would falsify Theorem 5.2.
4. Shell values. The shell values are $\{0, 1, 1.5, 2, 3\}$. A different set would falsify Theorem 6.1.
5. Shell assignment. Every mode's shell is determined by its $N/4$ label. A mismatch on any mode would falsify Theorem 7.2.
6. Equivalence. The two shell rules agree on all fourteen modes. A mismatch would falsify Theorem 7.4.

10 Conclusion

This paper has presented the shell assignment theorem of the Hodge Complex Standard Model. The shell assignment is determined by the $N/4$ labels through a step function with framework-constant thresholds. The kernel projector of the winding generator has orbit traces $(1, 6, 3)$ that generate the thresholds. The shell values are framework constants. The support-based rule and the $N/4$ step function agree on all fourteen modes.

We corrected a numerical error in the previous revision: the framework constant is $V = (d-1) \cdot 2^d = 48$, not $(d-1)^2 d = 36$.

The shell assignment is a theorem of the substrate, given the $N/4$ labels as inputs. The $N/4$ labels themselves are derived in HCSM-23 Rev. 8, Theorem 4.1.

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